

1)

a)

Name	Leading far field potential $V(r)$	Leading far-field $E(r)$
Monopole	$V \propto 1/r$	$E \propto 1/r^2$
Dipole	$V \propto 1/r^2$	$E \propto 1/r^3$
Quadrupole	$V \propto 1/r^3$	$E \propto 1/r^4$
Octuple	$V \propto 1/r^4$	$E \propto E \propto 1/r^5$

b)

The conditions on r under which the pure dipole formulas for V and $\mathbf{E}$ are valid approximations to the physical dipole is for points $r \gg d$ or by squeezing the charges very close together.

c)

For the "pure" dipole, the field lines will go straight out to the side because there is no space between the charges for there to be field lines between, while the "physical" dipole has enough space for there to be an E-field between the charges.

2)

As in example 4.1, we assume that electric cloud has the same shape, the nucleus moves a distance d and the external and internal electric fields are equal. $\Rightarrow$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{qd}{r^3} \Rightarrow qd = 4\pi\epsilon_0 r^3 E \quad |\text{At equilibrium \& } p = qd$$
$$\Rightarrow \alpha = 4\pi\epsilon_0 r^3 \Rightarrow \frac{\alpha}{4\pi\epsilon_0} = r^3$$

Where r is the radii of the atom. Can then calculate the polarizability of the atoms below:

Atom = radius [pm] $\Rightarrow$ polarizability [m^3]

Ag = 165 $\Rightarrow$ 4.49

He = 53 $\Rightarrow$ 0.0298

Fe = 156 $\Rightarrow$ 3.8

Cs = 298 $\Rightarrow$ 26.5

Ca = 194 $\Rightarrow$ 7.3

$$\text{Li} = 167 \Rightarrow 4.66$$

$$\text{Na} = 190 \Rightarrow 6.86$$

Can clearly see that the lower groups with higher period have much larger atomic polarizability and that the findings differ by a bit from the table in the book!

3)

a)

The dipole moment increasing with E makes physically sense when looking at smaller E -fields as the stronger the field is, the more the core charge of the atom is gonna move in the direction of the E -field until the internal E -field reaches equilibrium with the external. The atomic polarizability α represent the atoms ability to be polarized, depending on the direction of the E -field. It measures how "easy" it is to influence the atom by external E -fields, like how noble gasses like He have really low values and reactive substances like Cs have a high value, as they will easily give away electrons.

b)

For larger volume atoms, the electric forces between the positive center charges and negative electrons are weaker as they are further apart. This in turns make the atoms more polarizable, as it takes a smaller E -field to create large dipole moments.

The displacement of the electron cloud retaining its spherical shape is reasonable as each electron would be effected very little by a small E -field compared to the force of the larger positive charge in the center of the atom, which again would be effected a greater amount compared to the electrons.

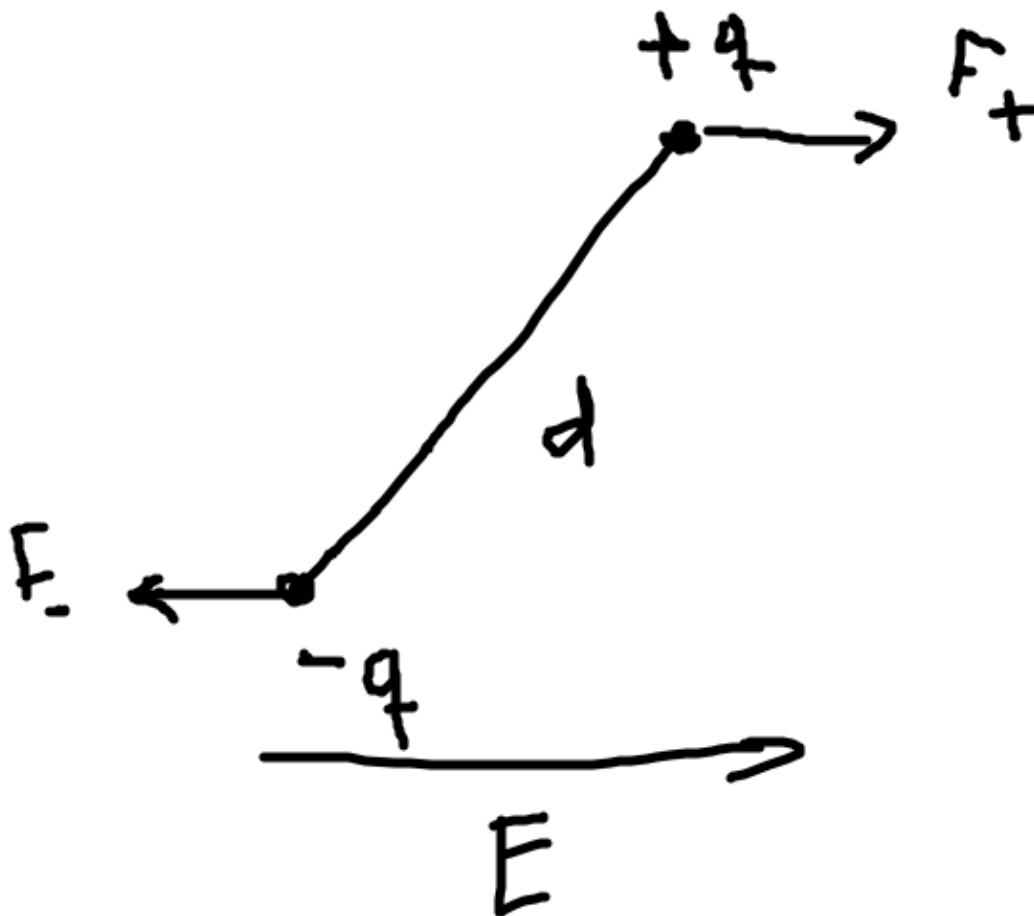
c)

For asymmetrical molecules like CO_2 , the orientation of the E -field and the internal electric between the atoms will both effect the induced dipole, making the direction dependent on the direction of the external field compared to the orientation of the molecule. Ex: The atomic polarizability may differ horizontally vs vertically across a molecule, making an E -field at an angle induce a non-parallel dipole moment.

d)

Given a dipole in an electric field, the forces the two charges experience will be equal but opposite directed, leading to a zero net force on the dipole. However the molecular torque will be equal to:

$$N = (r_+ \times F_+) + (r_- \times F_-) = \left[\left(\frac{d}{2} \right) \times (qE) \right] + \left[\left(\frac{-d}{2} \right) \times (-qE) \right] = qd \times E = p \times E$$



4)

a)

$W = Fd$ and $F = Eq$ and $qV = W$. By substituting and solving for E we get: $E = \frac{V}{d}$
 The electric field is then:

$$E = \frac{500}{0.001} = 500 \text{ kV/m}$$

b)

$$p = \alpha E = 0.66 \cdot 10^{-30} \cdot 4\pi\epsilon_0 \cdot 500000 = 3.67 \cdot 10^{-35} \text{ Cm}$$

c)

$$d = \frac{p}{e} = \frac{3.67 \cdot 10^{-35}}{1.60217663 \cdot 10^{-19}} = 2.29 \cdot 10^{-16} \text{ m}$$

$$\frac{d}{a_0} = 4.581 \cdot 10^{-6}$$

d)

Solving everything again for V with $d = a_0$ I got:

→ $E(V) := 1000 \text{ V}$

$$\rightarrow p(V) := \frac{5841}{2500} V^\pi$$

$$\rightarrow d(V) := \frac{5841}{4005441575000000000000} V \pi$$

NLØS: $\{V = 109139843.5075\}$

a)

b)

c)

$$\implies E(s)_{inside} = \frac{sp_b}{2\epsilon_0} \hat{s} = -\frac{sk}{\epsilon_0} \hat{s}$$